AETHER AI PHYSICAL SMART SPEAKER APPLIANCE

Commercial Product Blueprint • Hardware Bill of Materials • MCP GitHub Knowledge Pipeline

Target SBC: Raspberry Pi Zero 2 W / Orange Pi / ESP32-S3	**Software Stack:** Python 3.11, FastAPI, SQLite, FastMCP	**AI Engine:** Google Gemini 2.5 Flash + Edge Neural TTS	**Hardware BOM:** ~Rs. 2,870 ($34 USD) per unit

1. Executive Summary & Core Value Proposition

Commercial smart speakers (Amazon Echo, Google Home) only provide generic internet web search results. They cannot converse with proprietary private repositories, company SOPs, or project documentation. **Aether AI Smart Speaker** bridges this gap by marrying an ultra-low-cost physical appliance with an open **Model Context Protocol (MCP)** engine connected to your **GitHub Knowledge Base**. When documentation is pushed to GitHub in Markdown, the physical speaker instantly learns the knowledge in real-time with **$0 database hosting fees** and **$0 cloud subscription lock-in**.

> **The Core Innovation:** Hands-free voice queries (Hindi/English/Hinglish) triggering real-time MCP tool calling to fetch and audibly explain private GitHub repository knowledge through a 3W physical speaker in under 2 seconds.

2. Product Lineup & Real-World Commercial Use Cases

Product 1: 'DevDesk AI' — Software Engineer & Founder Desk Companion

A cylindrical desktop appliance next to a workstation monitor for hands-free codebase and architecture questions while coding.

- **Target Audience:** Software Engineers, DevOps Engineers, Technical Founders, Freelancers.
- **Voice Query:** *'Aether, what is the payment timeout configured in our repo webhook?'*
- **Speaker Output:** *'According to settings.py in main branch, payment timeout is 15 seconds with 3 retries.'*

Product 2: 'OfficeBrain Kiosk' — B2B Internal Company Smart Speaker

Positioned in office meeting rooms, pantries, or reception areas to resolve employee HR policy, onboarding, and SOP questions.

- **Target Audience:** Small & Medium Businesses (SMBs), Startups, Law Firms, Clinics, Coworking Spaces.
- **Voice Query:** *'Sick leave apply karne ka procedure kya hai aur approval kaun deta hai?'*
- **Speaker Output:** *'HR portal par apply karein; 2 din se zyada ke liye medical certificate required hai, aur direct manager approve karta hai.'*

Product 3: 'Showroom & Kiosk Assistant' — Interactive Product Demonstrator

Placed beside machines or retail demo displays to explain technical specs, warranty terms, and usage instructions directly from Git docs.

- **Voice Query:** *'Is machine ka power consumption kitna hai aur warranty kitne saal ki hai?'*

Product 4: 'Personal Second Brain' — Obsidian & GitHub Notes Companion

A bedside or home office speaker that queries private markdown journal notes, book summaries, and daily goals on command.

3. Technical Architecture & End-to-End Data Pipeline

The appliance runs a fully localized asynchronous pipeline engineered for sub-second response times, barge-in interruption, and offline persistence:

Pipeline Stage	Component / Technology	Technical Function
1. Audio Capture	INMP441 I2S MEMS Mic	Direct 24-bit digital capture at 16kHz PCM; eliminates analog hissing and ground loops.
2. Voice Trigger / Button	GPIO 17 / Keyword Spotter	Single-tap tactile hardware button or continuous VAD voice trigger initiates listening.
3. Speech-to-Text (STT)	Web Speech / Groq Whisper	Ultra-fast speech-to-text with multilingual Hindi, Hinglish, and English recognition.
4. Local Command Router	Deterministic Regex Engine	Evaluates Stop, Mute, Volume, Clear locally in <5ms ($0 API cost; bypasses cloud).
5. AI Brain & Tool Decider	Google Gemini 2.5 Flash	Evaluates user intent. Emits an MCP tool call if query pertains to the GitHub repository.
6. MCP Tool Execution	GitHub REST / FastMCP Client	Fetches target markdown doc from repository; feeds retrieved context back to Gemini.
7. Neural Voice (TTS)	Microsoft Edge Neural TTS	Synthesizes human-like speech via hi-IN-Swara, en-IN-Neerja, or en-US-Ava ($0 Free).
8. Audio Output & Amp	MAX98357A I2S Amp + 3W Spk	Class-D digital amplifier driving a 4Ω 3W speaker with instant barge-in cut-off.
9. Data Persistence	Embedded SQLite 3 (WAL)	Saves conversation history, volume, and latency metrics locally on flash ($0 fee).

4. MCP GitHub Knowledge Base Integration Mechanics

Aether integrates a dedicated FastMCP Client (backend/mcp/github_kb.py) exposing standard tool declarations to Gemini:

- **Tool 1: search_github_knowledge(query)** — Searches markdown files and documentation directories across your GitHub repository using fuzzy index and semantic keyword matching.
- **Tool 2: read_github_document(file_path)** — Fetches exact file contents via GitHub REST API with an intelligent local RAM cache to avoid redundant API hits.
- **Tool 3: list_knowledge_topics()** — Returns a summarized catalog of all indexed documentation files currently in the repository.

Zero Sync Overhead: Whenever an engineer or manager pushes an update (git push origin main), the physical speaker immediately reflects the updated facts upon the very next voice query.

5. Hardware Bill of Materials (BOM) & Unit Costing

Complete component shopping list for manufacturing one standalone physical smart speaker. All parts are standard and in stock across Indian robotics suppliers (Robu.in, Robocraze, ElectronicsComp) and Amazon:

Component	Recommended Model	Price (INR)	Price (USD)	Sourcing & Function Notes
1. Single Board Computer	Raspberry Pi Zero 2 W (Header Pin)	Rs. 1,650	$19.50	Quad-core 64-bit ARM, Wi-Fi 4, runs Linux/Python
2. Digital Microphone	INMP441 I2S MEMS Mic Module	Rs. 140	$1.70	Ultra-low noise digital 24-bit audio capture
3. Audio DAC Amplifier	MAX98357A I2S Mono 3W Amp	Rs. 160	$1.90	Class-D digital amplifier; high power efficiency
4. Loudspeaker	4 Ohm (4Ω) / 3W 40mm Driver	Rs. 85	$1.00	Optimized for vocal clarity and crisp speech
5. Visual Status Ring	WS2812B NeoPixel 8-LED Ring	Rs. 130	$1.55	RGB state indicator (Blue, Violet, Green, Red)
6. Tactile Push Button	12mm Tactile Button + Cap	Rs. 15	$0.20	Tap-to-Talk, Cancel audio, and Mic Mute
7. Flash Memory Storage	SanDisk 32GB Class 10 MicroSD	Rs. 280	$3.35	Houses OS, Python virtualenv & SQLite DB
8. Connecting Jumpers	Female-to-Female Wires (15 pcs)	Rs. 40	$0.50	Pin-to-pin wiring (No soldering required for prototype)
9. Power Supply Adapter	5V 2.5A Micro-USB Supply	Rs. 190	$2.25	Stable continuous 24/7 electrical feed
10. Enclosure Chassis	3D-Printed Cylindrical Box	Rs. 180	$2.15	Acoustic speaker cavity + mic port + LED top diffuser
TOTAL PROTOTYPE BOM	**Complete Turnkey Unit**	**Rs. 2,870**	**$34.10**	**Fully Functional Hardware Smart Speaker**

6. Hardware Pinout & Wiring Connection Map

The INMP441 Microphone and MAX98357A Amplifier share the common I2S audio bus clock lines on the Pi Zero 2 W 40-pin header:

Peripheral Module	Module Pin	Pi Zero 2W Pin	Signal / Function
INMP441 (Microphone)	VDD	Pin 1 (3.3V)	Power Supply (3.3V Logic ONLY - Never connect 5V)
INMP441 (Microphone)	GND & L/R	Pin 9 (GND)	Ground / Left Channel Audio Select
INMP441 (Microphone)	SCK	Pin 12 (GPIO 18)	I2S Bit Clock (Shared BCLK)
INMP441 (Microphone)	WS	Pin 35 (GPIO 19)	I2S Word Select / LR Clock (Shared LRCK)
INMP441 (Microphone)	SD	Pin 38 (GPIO 20)	I2S Data Input (Microphone PCM stream)
MAX98357A (Amplifier)	Vin	Pin 2 (5V)	5V High-Output Power Supply
MAX98357A (Amplifier)	GND	Pin 6 (GND)	Power Ground
MAX98357A (Amplifier)	BCLK	Pin 12 (GPIO 18)	I2S Bit Clock (Shared with Mic)
MAX98357A (Amplifier)	LRC	Pin 35 (GPIO 19)	I2S Word Select (Shared with Mic)
MAX98357A (Amplifier)	DIN	Pin 40 (GPIO 21)	I2S Audio Output to Speaker
Tactile Push Button	Pin 1 & 2	Pin 11 (GPIO 17) & GND	Active-Low Tap-to-Talk / Mute trigger
WS2812B LED Ring	5V, GND, DIN	Pin 4 (5V), Pin 20, Pin 32	PWM Controlled NeoPixel State Ring (GPIO 12)

7. Operational Cloud & API Costs (Zero-Cost Blueprint)

Aether AI is engineered for zero-to-negligible operating costs by leveraging free-tier APIs and local embedded compute:

Service / Resource	Provider & Model	Free Tier Limit	Monthly Cost (INR / USD)
AI Brain (LLM)	Google Gemini 2.5 Flash	15 RPM / 1M TPM 100% Free	Rs. 0 (Free) or ~Rs. 80/mo at scale
Speech-to-Text (STT)	Browser Web Speech / Groq	Unlimited / Free Tier	Rs. 0 (100% Free)
Neural TTS Voice	Microsoft Edge Neural TTS	Unlimited Multilingual	Rs. 0 (100% Free)
Knowledge Base Tool	GitHub REST API / MCP	5,000 requests/hour free	Rs. 0 (100% Free)
Database Persistence	Embedded SQLite 3 (WAL)	Local Flash Storage	Rs. 0 (100% Free)
Appliance OS & Services	Raspberry Pi OS (Bookworm)	Open Source Linux	Rs. 0 (100% Free)
TOTAL MONTHLY RUN COST	**1,000 Voice Queries / Month**	**Standard Daily Usage**	**Rs. 0 – Rs. 80 / mo ($0 – $1.00)**

8. Commercial Pricing Strategy & Profit Margins

Commercialization economics when packaging and selling Aether Smart Speaker as a finished hardware appliance:

Product Edition	Hardware BOM	Retail Price	Gross Margin (INR)	Profit Margin (%)
DevDesk AI (DIY Maker Kit)	Rs. 2,870	Rs. 5,999 ($72)	Rs. 3,129 per unit	**52.1%**
DevDesk AI (Assembled Box)	Rs. 3,100	Rs. 7,999 ($95)	Rs. 4,899 per unit	**61.2%**
OfficeBrain B2B Appliance	Rs. 3,500	Rs. 14,999 ($178)	Rs. 11,499 per unit	**76.6%**

9. Production Roadmap & Turnkey Deployment Steps

Step 1: Local Verification on Windows (Current Phase): Validate Gemini function calling with your GitHub MCP tools and SQLite session persistence locally.

Step 2: Component Procurement: Order Raspberry Pi Zero 2 W, INMP441, MAX98357A, and 4Ω 3W mini speaker from Robu.in / Robocraze (~Rs. 2,870).

Step 3: Hardware Assembly & Jumper Pinning: Connect the 10 jumper wires according to Section 6 pinout map; snap into a 3D-printed cylindrical enclosure.

Step 4: 1-Command Automated Installation: Flash Raspberry Pi OS (Bookworm 64-bit Lite) and execute bash scripts/setup_appliance.sh. The system registers a systemd auto-boot daemon.

Step 5: GitHub Knowledge Base Linkage: Add markdown documentation into your GitHub repository. The speaker instantly answers queries in natural voice!